Machine Learning for Real-Time Diagnostics of Cold Atmospheric Plasma Sources

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***Abstract*—Real-time diagnostics of cold atmospheric plasma (CAP) sources can be challenging due to the requirement for expensive equipment and complicated analysis. Data analytics that rely on machine learning (ML) methods can help address this challenge. In this paper, we demonstrate the application of several ML methods for real-time diagnosis of CAPs using information-rich optical emission spectra and electro-acoustic emission. We show that data analytics based on ML can provide a simple and effective means for estimation of operation-relevant parameters such as rotational and vibrational temperature and substrate characteristic in real-time. Our findings indicate a great potential promise for ML for real-time diagnostics of CAPs.**

***Index Terms*—Cold atmospheric plasma (CAP), electro-acoustic signal, Gaussian process (GP), *k*-means clustering, linear regression, machine learning (ML), optical emission spectrum (OES), real-time diagnostics.**

I. Introduction

COLD atmospheric plasma (CAP) sources have been recently explored for a variety of medical applications [1]–[3], [59]. However, CAP sources can suffer from unmonitored variabilities in operation. For example, atmospheric pressure plasma jets (APPJs) exhibit long timescale drifts [4], sharp gradients in temperature and species concentrations [5], and high sensitivity to disturbances such as ambient humidity and substrate impedance [6], [7]. Mode transitions are also frequently observed in CAP sources, e.g., streamer-to-spark transitions in corona-like air discharges [8]. Such variabilities pose a significant challenge to CAP research and applications, particularly where plasma interacts with complex systems, as in plasma medicine [9], [10]. Thus, monitoring variabilities in plasma characteristics in real-time can be especially useful for both understanding and minimizing irreproducible plasma effects. Real-time diagnostics can allow detection of abnormal or undesirable operating conditions and drifts in key plasma characteristics. Furthermore, real-time plasma diagnostics are indispensable for advanced process control in order to (partly) mitigate variabilities of CAP sources and improve their operational reliability [11]–[14].

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Direct quantitative diagnostics of CAP sources pose a considerable challenge. Methods such as laser-induced fluorescence (LIF) [15], mass spectrometry [16], and spontaneous Raman scattering [17], for example, rely on sophisticated instrumentation and specially designed experimental setups. This is in stark contrast with the current practice of CAP operation in plasma medicine that relies on the flexibility of hand-held treatment in the absence of plasma diagnostics [18]. Spectral information from various sources such as optical and electro-acoustic emission can be used for plasma diagnostics. Emission signals are often easy to acquire and typically contain a wealth of implicit information about the plasma characteristics [19]. However, this information is often indirect and requires computationally expensive analysis to extract physical quantities such as gas and electron temperatures or reactive species concentrations [20]. This can make the use of spectral information for real-time diagnostics impractical.

Investigation of spectral information for real-time diagnosis of CAP sources has received some attention in the literature. Most notably, O'Connor *et al.* [21] reported on a multivariate method based on principle component analysis for correlating various optical emission spectrum (OES) peaks to electrical properties and electron density for a dielectric barrier discharge in helium. In contrast to the well-established method of analyzing OES by generating and fitting synthetic spectra to measurements [19], [22], these authors note the potential for developing correlations across large data sets for novel applications.

Electro-acoustic emission also contains useful information on plasma characteristics. Diagnostics based on electro-acoustic emission is commonly used in high-temperature materials processing applications such as arc welding [23] and plasma anodizing [24]. Law *et al.* [25] investigated the application of electro-acoustic emission for process monitoring and control in a DC pulse-modulated APPJ in air. These authors point to the usefulness of spectrally resolved electro-acoustic emission as a diagnostic tool for various plasma characteristics including surface properties, dissipated power, and separation distance between jet nozzle and substrate. O'Connor and Daniels [26] investigated the electro-acoustic emission of a nonthermal helium plasma jet for process monitoring. They utilized wavelet transforms of electro-acoustic emission for detection of flow modes and operational anomalies such as sparking phenomena.

In addition, the voltage-current signal is easily available in most plasma setups. Utilization of these signals for plasma diagnostics has been investigated by Walsh *et al.* [27] in the context of mode behavior of a nonthermal kHz-excited APPJ in helium. These authors demonstrate that phase-space representations of current signals can help distinguish between the three identified modes, although later work by Liu and Kong [28] indicates the detection of mode behavior may be more complicated in the presence of humidity. Law *et al.* [29] reviewed potential application of electro-acoustic emission and current signals for real-time diagnostics and process control.

Data analytics can present an alternative to the commonly used purely physics-based approaches for extracting information from spectra. With the increasing size of available data sets, algorithms for discovering patterns in data, broadly referred to as machine learning (ML) methods, have found success as data analytics tools [30], [31]. Some notable applications of ML for data analytics, among many others, include gene discovery [32], remote sensing [33], meaning and sentiment analysis of text [34], [35], object recognition in images [36], as well as real-time diagnostics and modeling of low-pressure plasma etch processes [37]–[41]. An overview of ML applications for modeling and simulation, diagnosis, and process control of nonequilibrium plasmas is discussed in [42]. The availability of easy-to-use software packages such as the open source package scikit-learn [43] and Google's Tensor Flow [44] has made ML methods more accessible, contributing to their popularity.

In this paper, we demonstrate the potential of data analytics for real-time diagnostics of CAP sources. Through use of ML, physical quantities, which are otherwise difficult to obtain in real-time due to limitations of instrumentation (e.g., optical setup required for LIF) or time required for analysis (e.g., fitting OES spectra), can be inferred by utilizing the raw spectral information. We present three case studies demonstrating the application of data analytics tools for real-time CAP diagnostics.

1) Determination of rotational and vibrational temperature from OES.
2) Discrimination between a conductive and an insulating substrate from OES.
3) Determination of the electrode-to-substrate separation distance from electro-acoustic emission.

In the rest of this paper, we first present an overview of selected ML methods, then describe the utilized experimental setups, and finally present results pertaining to each of the three investigated cases.

II. Machine Learning Methods

In this section, we review the ML methods used in this paper (see [30], [31], [45], [46] for a detailed description). The selection of a suitable ML method relies on two principal considerations related to the output data: 1) whether the output variables are continuous or discrete and 2) whether they are used for training a model. When an ML method uses the output data for training, it is known as *supervised* learning, where the goal is to predict continuous (e.g., *regression*) or discrete (e.g., *classification*) output variables. On the other hand, *unsupervised* learning only uses the input data to discover patterns in the data. For example, clustering methods are used to separate data into discrete bins.

Data sets used for ML consist of inputs denoted by $X \in \mathcal{R}^{N\times n}$ and, where applicable, outputs denoted by $Y \in \mathcal{R}^{N\times m}$. Here, n and m are the dimensions of the inputs and outputs, respectively, and N is the number of samples in the data set. We denote one sample of inputs by $\mathbf{x} \in \mathcal{R}^n$ and one sample of outputs by $\mathbf{y} \in \mathcal{R}^m$. We further denote predictions of an ML model by $\hat{Y} \in \mathcal{R}^{N\times m}$ and a single instance of prediction by $\hat{\mathbf{y}} \in \mathcal{R}^m$. Data are typically divided into *training*, *validation*, and *test* sets. In general, the majority of the data set (50%–90%) is used for training and validation [31]. During *training*, mathematical descriptions of the relationships among the data are inferred. Training typically involves a parametric or nonparametric fitting process. In order to systematically assess the quality of inferences made by an ML model, a *validation* procedure is employed. A common approach is k-fold cross-validation procedure in which the training data are partitioned into k complementary training and validation subsets or *folds* [30]. $k-1$ of the subsets are used for training and the remaining subset is used for validation. To overcome the inherent randomness of arbitrarily partitioning a data set, this process is systematically repeated for different partitioned sets [30]. Furthermore, ML models often have some *hyperparameters*, whose values cannot be directly estimated from data and thus are fixed prior to training. The models are often trained using different values of hyperparameters and subsequently the appropriate values are selected based on the model performance in the validation step. Finally, in the *testing* stage, the predictive capability of the ML model is assessed against an independent data set. In this paper, the performance of the ML model is quantified in terms of some error metrics such as root mean squared error (RMSE) and error fraction when the output variables are continuous and discrete, respectively. These error metrics are defined as

$$\mathrm{RMSE}\left(Y, \hat{Y}\right) = \frac{1}{N}\sum_{i=1}^{N}\sqrt{\left(\hat{\mathbf{y}}_i - \mathbf{y}_i\right)^2} \tag{1}$$

and

$$\text{Error Fraction}\left(Y, \hat{Y}\right) = \frac{1}{N}\sum_{i=1}^{N}\mathbb{1}\left(\hat{\mathbf{y}}_i \neq \mathbf{y}_i\right). \tag{2}$$

In expression (2), the indicator function $\mathbb{1}$ takes the value 1 when the condition $\hat{\mathbf{y}}_i \neq \mathbf{y}_i$ is met and 0 otherwise.

In Section IV, we explore the use of supervised learning for determination of rotational and vibrational temperatures (linear regression) and the electrode-to-substrate separation distance (Gaussian process (GP) regression), and unsupervised learning for discrimination between glass and metal substrates (k-means clustering). An overview of the adopted ML methods is presented in the remainder of this section.

A. Linear Regression

Linear regression is widely used in statistical analysis [47]. In linear regression, output variables are described by a linear combination of some (nonlinear) function of the inputs [30]. For a scalar output, a regression model can be written as

$$\hat{y} = \sum_{j=1}^{O} w_j \phi_j(\mathbf{x}) \tag{3}$$

where $\mathbf{w} = [w_1, \ldots, w_O]$ is the vector of weights, $\mathbf{x}$ is the vector of inputs, and $\phi_j(\mathbf{x})$ denotes basis functions. Note that expression (3) readily generalizes to higher output dimensions. A common choice for the basis functions, which is utilized in this paper, is powers of the inputs defined as

$$\phi_j(\mathbf{x}) = \mathbf{x}^j. \tag{4}$$

In this case, O, the highest order of the polynomial basis functions in (4), comprises the *order* of the regression model. We treat O as a hyperparameter.

In linear regression, the weights $\mathbf{w}$ are determined by solving a least-squares minimization problem. A key challenge in regression is over-fitting. In particular, when a large model order O in (3) is selected, a large number of weights should be fitted. This can lead to a regression model that describes the data too closely, even describing the random noise in the data. To avoid over-fitting, regularized least-squares estimation is commonly used. Here, we utilize the least absolute shrinkage and selection operator (LASSO) method, where the regression problem for a scalar output is expressed as [48]

$$\min_{\mathbf{w}} \left\| \hat{y} - y \right\|_2^2 + \alpha \|\mathbf{w}\|_1 \tag{5}$$

with α being the regularization hyperparameter. In classic linear regression, α is set to 0, leaving only the first term in the least squares problem (5). The selection of α governs what is known as the *bias-variance tradeoff* [30]. An excessively large value of α gives rise to *under-fitting* of the data, therefore leading to predictions with a large bias and a small variance. On the other hand, excessively small values of α cause *over-fitting* of the data, yielding predictions with small bias but high variance, since the regression model also captures the noise of the data [46]. In this paper, we select the appropriate value of α during the validation step, as described in Section IV-A.

The RMSE metric (1) provides a measure of the average error between true values and predictions; however, this metric is not sufficiently sensitive to noise in the measurement. We therefore also use the R^2 score, which indicates the capability of the regression model in capturing the variability in the output variables. The R^2 score is defined by

$$R^2 = 1 - \frac{\sum_{i=1}^{N} \left(y_i - \hat{y}_i\right)^2}{\sum_{i=1}^{N} \left(\bar{y} - \hat{y}_i\right)^2} \tag{6}$$

where N is the total number of samples, $\hat{y}$ is computed from (3), y_i is the ith measured sample of the scalar output y, and $\bar{y}$ is the mean of the true values of the scalar output

$$\bar{y} = \frac{1}{N} \sum_{i=1}^{N} y_i. \tag{7}$$

B. k-Means Clustering

k-means clustering is an unsupervised ML method used to separate data into k distinct groups or clusters [49]. The clusters are characterized in terms of their center points or *centroids*. In k-means clustering, the centroids of the clusters are initialized at random points and the distance between each input in a cluster to the corresponding centroid is evaluated in terms of some distance metric such as the Euclidean norm. The clusters are then iteratively updated and input clustering is refined by minimizing the sum of distances between the centroids and the inputs in each cluster. k-means clustering relies on the minimization problem

$$\min_{\boldsymbol{\mathcal{S}}} \sum_{i=1}^{k} \sum_{\mathbf{x} \in \mathcal{S}_i} \left\| \mathbf{x} - \boldsymbol{\mu}_i \right\|_2^2 \tag{8}$$

where $\mathcal{S}_i$ is the set corresponding to the ith cluster, $\boldsymbol{\mathcal{S}} \in \mathcal{R}^n$ is the set of $\mathcal{S}_i$, and $\boldsymbol{\mu}_i \in \mathcal{R}^n$ denotes the centroid of the cluster i. The user-defined hyperparameter in k-means clustering is the value of k. The choice of k can depend on some prior knowledge of data, or can rely on heuristics. In the case study presented in Section IV-B, we utilize prior knowledge of data for selection of k.

Note that since k-means clustering is an unsupervised learning method it does not predict clusters of new samples [46]. Nevertheless, new samples can be assigned to a cluster either by precomputing the relevant groups and dividing the space into regions, or by retraining the k-means clustering model when new data becomes available.

C. Gaussian Process Regression

GP regression is a nonparametric supervised ML method that relies on a probabilistic interpretation of the data [50]. In GP regression models, the outputs are assumed to have a joint Gaussian distribution [31]. Thus, the GP regression problem is formulated as estimation of the probability distribution of the predicted variables *conditional* on the training data. The joint *prior* probability distribution of the outputs is expressed as

$$\begin{pmatrix} Y \\ \hat{Y} \end{pmatrix} \sim \mathcal{N}\left(\mathbf{0}, \Sigma\right) \tag{9}$$

where $\mathcal{N}$ represents a Gaussian distribution with mean $\mathbf{0}$ and covariance matrix Σ. Without loss of generality, the mean of the joint distribution can be assumed to be zero. Even if the outputs are distributed around some nonzero mean, this value can be subtracted from the joint distribution to satisfy the zero-mean assumption. The covariance matrix Σ is defined by

$$\Sigma = \begin{pmatrix} K & K_* \\ K_*^\top & K_{**} \end{pmatrix} \tag{10}$$

where K, K_*, and K_{**} denote the individual covariance matrices corresponding to combinations of training and test data sets: $K = K(X_{\text{train}}, X_{\text{train}})$, $K_* = K(X_{\text{train}}, X_{\text{test}})$, and $K_{**} = K(X_{\text{test}}, X_{\text{test}})$. The covariance matrices K can be defined in terms of a positive definite kernel function

$$K\left(\mathbf{x}, \mathbf{x}'\right) = -\lambda \exp\left(\frac{1}{2\sigma^2} \left\| \mathbf{x} - \mathbf{x}' \right\|_2^2\right) \tag{11}$$

where $\mathbf{x}$ and $\mathbf{x}'$ are inputs pairs belonging to appropriate data sets, and σ and λ are the model hyperparameters. The kernel function K describes a measure of distance or similarity between the input pairs. Given the prior Gaussian distribution (9), the goal in GP regression is to predict the *posterior* distribution of the test data *conditional* on the training data.

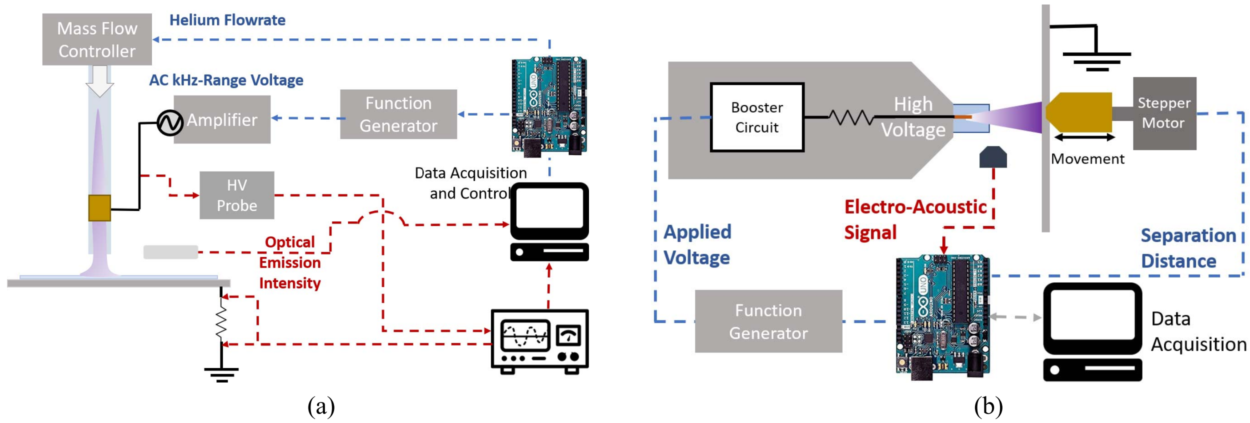

Fig. 1. Schematics of the (a) kHz-excited APPJ in helium and (b) plasma flashlight. Dashed lines indicate the flow of information, where blue represents actuation and red represents measurements.

The expected value $\mathbb{E}$ and covariance $\hat{\Sigma}$ of this posterior distribution can be computed using Bayes' rule [50]

$$\mathbb{E}\left(\hat{Y}|Y, X_{\text{train}}, X_{\text{test}}\right) = K_* K^{-1} Y \tag{12}$$

and

$$\hat{\Sigma}\left(\hat{Y}|Y, X_{\text{train}}, X_{\text{test}}\right) = K_{**} - K_* K^{-1} K_*. \tag{13}$$

Expressions (12) and (13) fully define the distribution of outputs predicted by a GP model.

A key difference between GP and linear regression is that a GP model does not have a parametric functional form. Instead, GP explicitly uses the provided training data when making predictions. The hyperparameters of the kernel function (11) can be obtained *a priori*, for example, by maximizing the log marginal likelihood function with respect to the hyperparameters [50]. The nonparametric nature of GP is advantageous in predicting arbitrarily nonlinear behaviors. Another important feature of GP regression is that it predicts a variance associated with the expected value of the prediction, which can be used to provide confidence bounds on the model predictions.

III. Experimental Methods

We utilize two different CAP sources to demonstrate the applicability of ML to different types of discharges; a kHz-range APPJ in helium and the "plasma flashlight" [51], a hand-held corona-like discharge in air. The kHz-range APPJ has the advantage of being a very stable discharge with easily obtainable OES spectra. On the other hand, the corona-like discharge produces clearly audible electro-acoustic emission.

The schematic of the two setups are presented in Fig. 1. The specifications of the APPJ [Fig. 1(a)] are similar to those presented in [12]; a quartz tube (ID = 3 mm and OD = 4 mm) serves the dual purpose of flow channel and dielectric barrier. Plasma is ignited by applying a sinusoidal voltage at a frequency of 20 kHz on a copper ring electrode wrapped around the tube. An aluminum plate at a fixed distance of 4 mm from the tube nozzle acts as ground as well as the conductive substrate. A borosilicate microscope cover slip is placed under the APPJ to be used as the dielectric substrate. The corona-like discharge [Fig. 1(b)] is identical to the plasma flashlight device reported in [51]. The device consists of a needle-type electrode situated in a plastic nozzle. The plasma is ignited by DC voltage amplified by booster circuit and exhibits self-pulsing behavior.

Fitting accurate ML models generally requires large amounts of data, which are both labor and time intensive to collect manually. Therefore, automation of the data acquisition and processing is a key requirement for effective implementation of ML algorithms. To this end, both setups are equipped with automated data acquisition and actuation systems based on the open-source micro controller Arduino UNO. A single board computer (Raspberry Pi 3) is used to coordinate actuation and data acquisition from various sources. Automated data collection and actuation is implemented in Python and the ML algorithms are implemented using the scikit-learn package [43].

In the APPJ setup [Fig. 1(a)], power and flow can be actuated within ranges of 1.5–5 W and 1–3 slm, respectively. The applied power is maintained via a proportional-integral (PI) controller, based on instantaneous measurements of current (via an 830 Ω resistor) and voltage (via Tektronix P6015A high voltage probe) signals. Flow is manipulated through a mass flow controller (UNIT UFC-1660) interfaced to Arduino UNO. The OES signal is collected with a spectrometer (OceanOptics USB2000+, 0.375-nm resolution). In the plasma flashlight setup [Fig. 1(b)], the separation distance is manipulated through a linear actuator based on a stepper motor controlled by the Arduino UNO. The electro-acoustic emission is collected in the range of 0.5–25 kHz with a resolution of 200 Hz using an adjustable gain microphone (MAX4466). The fast Fourier transform of the signal is implemented via the Open Music Library [52].

IV. Results and Discussion

We now discuss the application of the ML methods of Section II for real-time diagnostics of the CAP discharges shown in Fig. 1 using three case studies.

1) Determination of rotational and vibrational temperatures of the APPJ using OES.
2) Discrimination between a conductive and an insulating substrate using OES.
3) Determination of the electrode-to-substrate seperation distance using electro-acoustic emission.

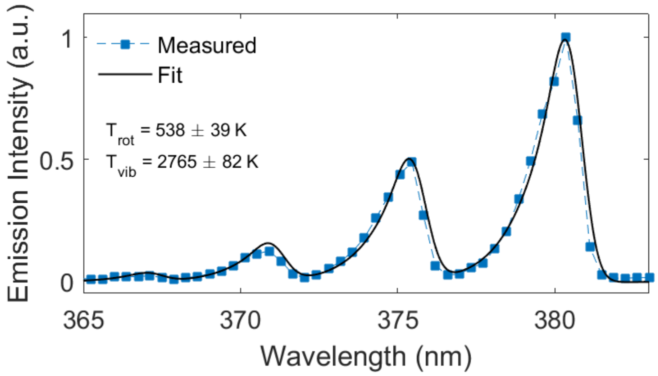

Fig. 2. Example OES spectrum of N_2 ($C^3\Pi_u \rightarrow B^3\Pi_g$) second positive transition and corresponding fit from Massive OES [22] used to estimate rotational and vibrational temperatures of the APPJ operated at $P = 3$ W and $q = 1.5$ slm over the glass substrate.

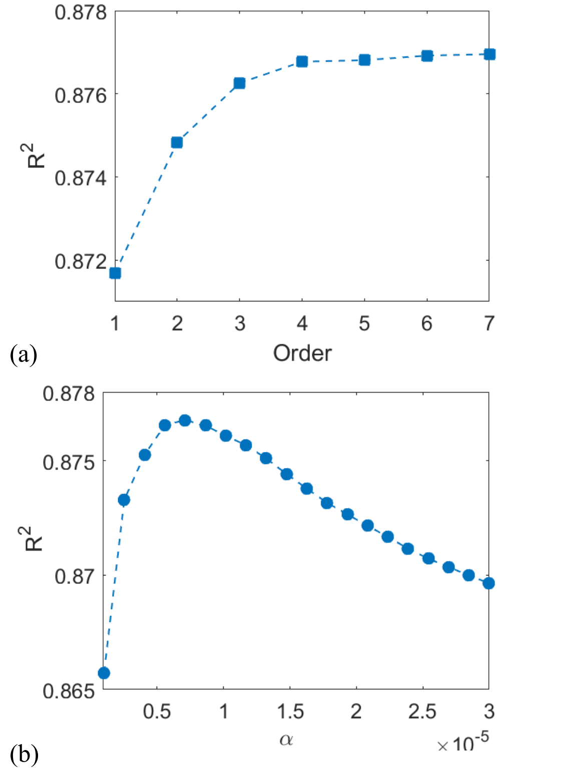

Fig. 3. Determination of rotational and vibrational temperatures using linear regression. Effect of (a) the order of the linear regression model and (b) the regularization parameter α on the predictive capability of the model as quantified by R^2 score.

A. Determination of Rotational and Vibrational Temperatures Using OES

Rotational (T_{rot}) and vibrational (T_{vib}) temperatures contain vital information about gas temperature and energy transport processes within the discharge. Different OES peaks such as OH ($A^2\Sigma^+ \rightarrow X^2\Pi$ at 306–328 nm) and N_2 ($C^3\Pi_u \rightarrow B^3\Pi_g$ at 365–390 nm) transitions are commonly used to estimate rotational and vibrational temperatures [19], [20]. The current practice is to estimate these temperatures offline using specialized software (e.g., SpecAir [19]) to achieve a fit between generated synthetic spectra and measured OES spectra. Due to the presence of a large number of fitting parameters, this approach can be computationally expensive for real-time diagnostics, taking several seconds to few minutes. Here, we propose a data-driven diagnostics tool based on linear regression (see Section II-A) for real-time inference of the rotational and vibrational temperatures from the OES spectra.

The regression model is trained using 1500 samples of the normalized peaks of N_2 second positive system between 365–385 nm (input) and estimated rotational and vibrational temperatures (outputs). A typical OES spectrum over the wavelengths corresponding to N_2 ($C^3\Pi_u \rightarrow B^3\Pi_g$) transition and the fit of the synthetic spectrum are shown in Fig. 2. The training data for values T_{rot} and T_{vib} are determined via fitting synthetic spectra in Massive OES [22]. The data set is collected over a range of operating conditions: varying He flow, applied power, and substrate type (i.e., insulating and conductive). The data are split into 80% training and 20% testing segments. The training data are presented in the Appendix.

We considered two hyperparameters for linear regression: 1) the model order as defined by the choice of basis functions (4) and 2) the choice of the regularization parameter α in (5). The model is validated using tenfold cross-validation. Figs. 3(a) and (b) show the average R^2 score of the model predictions over ten folds as a function of the model order and as a function of the regularization parameter α, respectively. We observe that the R^2 score increases with model order and appears to plateau at an order of four [Fig. 3(a)]. Above this model order, increasing the complexity of the model does not provide further improvement in the predictive capability of the model. In contrast, we observe that the R^2 score is low for low values of α, possibly causing the model to over-fit [Fig. 3(b)]. In this case, the model is unnecessarily complex due to fitting the noise in the output data. When α is increased beyond $\alpha = 0.7 \times 10^{-5}$, the model tends to under-fit. Thus, we choose the model order $O = 3$ in (3) and the regularization parameter $\alpha = 0.7 \times 10^{-5}$ in (5) in order to strike a balance in the bias-variance tradeoff.

Fig. 4 depicts the predictions of the linear regression model against the test data. The operating conditions under which the test data are obtained are shown in Fig. 4(b). The predictive capability of the linear regression model is quantified separately for the two outputs T_{rot} and T_{vib}. The RMSE values, calculated using (1), are 30.2 K and 141.0 K, respectively, and the R^2 scores are computed as 0.79 for both outputs. Fig. 4(a) suggests that the regression model can describe the fitted T_{rot} and T_{vib} fairly well. A notably poor performance is observed between sampling instants 210–250, where the flow rate is decreased and power is increased on the conductive substrate. Under these conditions, the fitted temperatures are subject to more noise. This can be attributed to the fact that the plasma jet under these conditions moves rapidly from point to point on the surface, disrupting the OES signal collection.

The results illustrated in Fig. 4 indicate that the predictions of the linear regression model are comparable to the fitted values from Massive OES under a range of operating conditions. Further testing also revealed that the linear regression model retains its accuracy even under additional variations

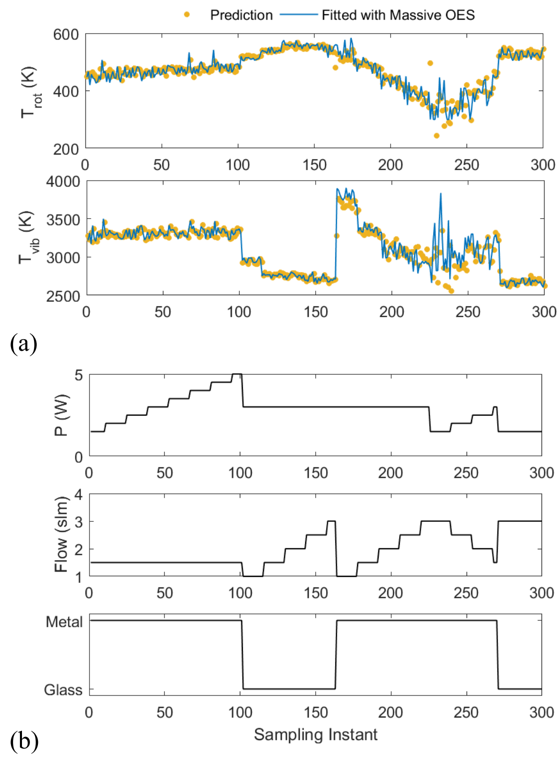

Fig. 4. Determination of rotational and vibrational temperatures using linear regression. (a) Predictions of the linear regression model for $T_{\rm rot}$ and $T_{\rm vib}$ compared against the fitted values from Massive OES using the test data and (b) operating conditions (applied power, He flow rate, and substrate type) under which the test data are obtained.

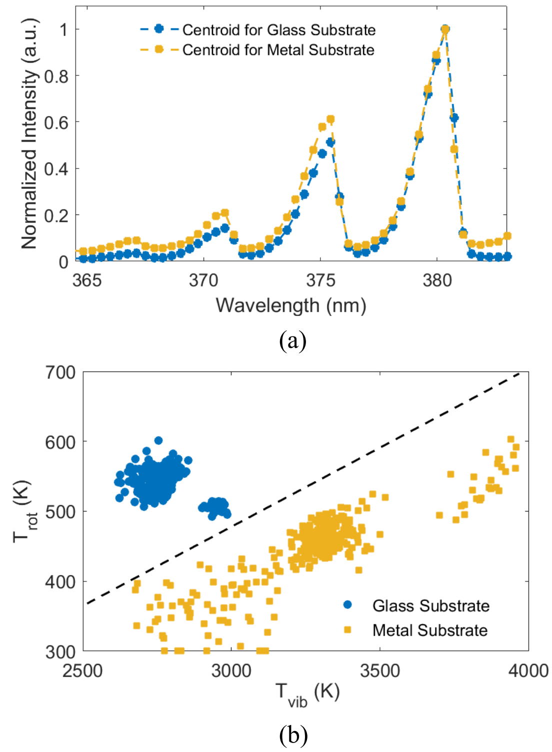

Fig. 5. Substrate discrimination using k-means. (a) Centroids of the clusters that correspond to glass and metal substrates. (b) Rotational temperature $T_{\rm rot}$ and vibrational temperature $T_{\rm vib}$ in the training data, fitted using Massive OES.

that may significantly impact the emission from the second positive transition of N_2, such as variations in the APPJ tip-to-substrate separation distance and O_2 admixture in the working gas (results not reported). The generalizability of the linear regression model can be attributed, in part, to the inherent repeatability of underlying physical phenomena in this type of discharges [53]. Thus, the linear regression model can provide an effective tool for real-time inference of $T_{\rm rot}$ and $T_{\rm vib}$. This type of regression model relating OES to physical quantities can be a powerful diagnostic tool even beyond the determination of temperatures. When additional quantitative measurements such as LIF and picosecond second harmonic generation are available for training, supervised regression models might be developed using OES data to provide real-time estimates of other process variables such as chemical species concentrations and electric field strength.

B. Substrate Discrimination Using OES

APPJ characteristics can significantly change as a function of the substrate properties such as the conductive or insulating nature of the target substrate [6]. However, it is often impractical to directly measure the substrate properties in real-time. Changes in the optical emission of the discharge can provide an indication of changing substrate properties and allow detection of distinct substrates, for example, distinguishing tumorous tissue from healthy tissue [54]. Here, we develop a real-time diagnostics tool based on k-means clustering to detect the substrate type based on OES measurements. We utilize the same training data set as in the previous section (see the Appendix). The data set consists of 1500 samples, 1000 of which are collected over a glass substrate and 500 are collected over a metal substrate. The inputs are the raw OES spectra of the second positive transition of N_2 as in the previous section. Labels of the substrate types are the outputs. However, as k-means clustering is an unsupervised learning method, the output labels are not used for training. We aim to cluster the OES spectra in two distinct groups ($k = 2$). We use tenfold cross-validation to validate the ability of the k-means clustering algorithm to assign data to the appropriate substrate labels. We find that the k-means clustering algorithm is able to classify the glass and metal substrates with an error fraction of 0.01. That is, the clustering algorithm misclassifies the substrate only 1% of the time.

The centroids, or the average spectra, corresponding to the two clusters are shown in Fig. 5(a). The k-means clustering algorithm clusters the OES spectra into two classes coinciding with the glass and metal substrates. We further verified the clustering with respect to the two substrate types by plotting the $T_{\rm rot}$ and $T_{\rm vib}$ values fitted from Massive OES. As shown in Fig. 5(b), the rotational and vibrational temperatures over glass and metal are clearly separated by the dashed line. The clustering of temperatures is not particularly surprising given the sensitivity of the plasma to electrical properties of the substrate. However, to the authors' knowledge, this clustering behavior has not been previously reported.

We tested the clustering algorithm for substrate discrimination in real-time. In this case, the glass cover slip used as

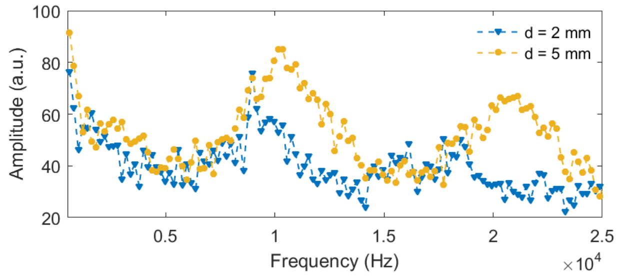

Fig. 6. The fast Fourier transform of the electro-acoustic emission of the plasma flashlight recorded at two different electrode-to-substrate separation distances of $d = 2$ mm and $d = 5$ mm.

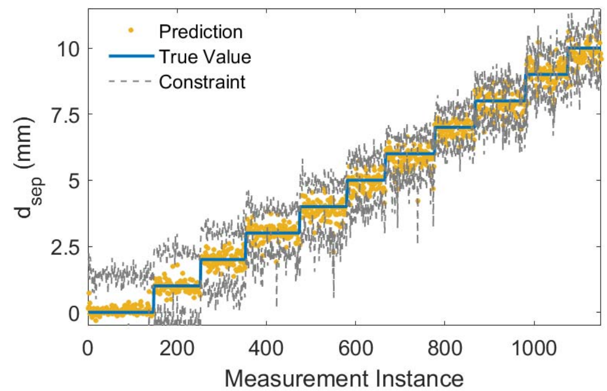

Fig. 7. Determination of the electrode-to-substrate separation distance of the plasma flashlight using GP regression. Comparison of predictions of the GP regression model against the test data. The confidence interval of the GP predictions, quantified as one standard deviation around the predicted mean, is shown with solid gray lines.

the insulating substrate (see Fig. 1(a)) was removed after 30 s of operation and was reinserted after 30 s. The APPJ was operated with the applied power of 3 W and He flow of 1.5 slm. Under these conditions, we observed perfect identification of the substrate type (an error fraction of 0), suggesting that the k-means clustering algorithm can be used for real-time diagnosis of the substrate type. We note that the case examined here was fairly simple, since the transition between a purely conductive and a purely insulating substrate is fairly drastic. Nevertheless, this case study demonstrates the promise of unsupervised clustering for real-time diagnostics of discrete phenomena. Knowledge of discrete phenomena, such as substrate type or plasma modes, can greatly increase the flexibility of operation, for example, for treatment of electrically heterogeneous substrates.

C. Determination of Separation Distance Using Electro-Acoustic Emission

Here, we use the hand-held battery-operated air discharge, the plasma flashlight, shown in Fig. 1(b). This device exemplifies typical plasma medical devices, such as KINPen (neoplas tools GmbH [55]), as it is hand-held and relies on the expertise of the user for reliable operation. In these devices, changes in the electrode-to-substrate distance can generate significant variability in operation due to the sharp gradients and drastic changes in the plasma properties [5]. Hence, real-time measurement of the separation distance can be useful. However, conventional noncontact sensors based on infrared [56], ultrasound [57], or time-of-flight [58] may not be able to readily address this issue as the desired measurement range is particularly small and the discharge interference with electronics can limit their use. Here, we aim to use the plasma discharge information to infer the electrode-to-substrate separation distance of the plasma flashlight in real-time.

The plasma flashlight has audible electro-acoustic emission, which noticeably changes with variations in the separation distance. Based on [25] and [26], we use the fast Fourier transform (FFT) to analyze the effect of the separation distance on the sound signal in the frequency domain. Fig. 6 shows an example of the FFTs of the electro-acoustic emission collected under two separation distances of 2 and 5 mm. We use GP regression to describe the complex, nonlinear relationship between the FFT of the electro-acoustic emission and the separation distance (see Section II-C). The data set for training and testing the GP regression model consists of 2500 samples collected over a range of labeled separation distances d_{sep}. Sixty percent of the data are used for training and validation while the rest is reserved for testing. The training data are given in the Appendix.

In the GP regression model, the hyperparameters in (11) are systematically chosen by maximizing the log marginal likelihood of predictions with respect to the hyperparameters [50]. This results in $\lambda = 1.73$ and $\sigma = 0.08$. After training and validation of the GP model, its performance is evaluated with respect to the test data, as shown in Fig. 7. The performance of the GP model is quantified with the R^2 score of 0.98 and RMSE of 0.34 mm. Fig. 7 suggests that the GP model reliably predicts the true value of the separation distance. The noisy nature of the prediction is attributed to the high noise level of the raw electro-acoustic emission data. Notably, the standard deviation predicted by the GP prediction is smaller for larger separation distances, as shown by the dashed lines in Fig. 7. Overall, the GP regression model is capable of predicting the separation distance based on electro-acoustic emission fairly accurately. Thus, the GP regression model provides a real-time diagnostic for determination of the separation distance, for example, to monitor if the separation distance is outside a prescribed region that is safe for plasma treatment.

V. Conclusion

This paper demonstrates three examples of machine learning-based data analytics for real-time diagnostics of atmospheric pressure plasmas.

1) Determination of rotational and vibrational temperatures from OES using linear regression.
2) Discrimination of different types of substrates from OES using k-means clustering.
3) Determination of the electrode-to-substrate separation distance from electro-acoustic emission using Gaussian process regression.

A common feature of all the investigated examples is that difficult-to-obtain information is extracted from spectral data that can be measured in real-time. This hinges on offline collection and processing of possibly large quantities of data to

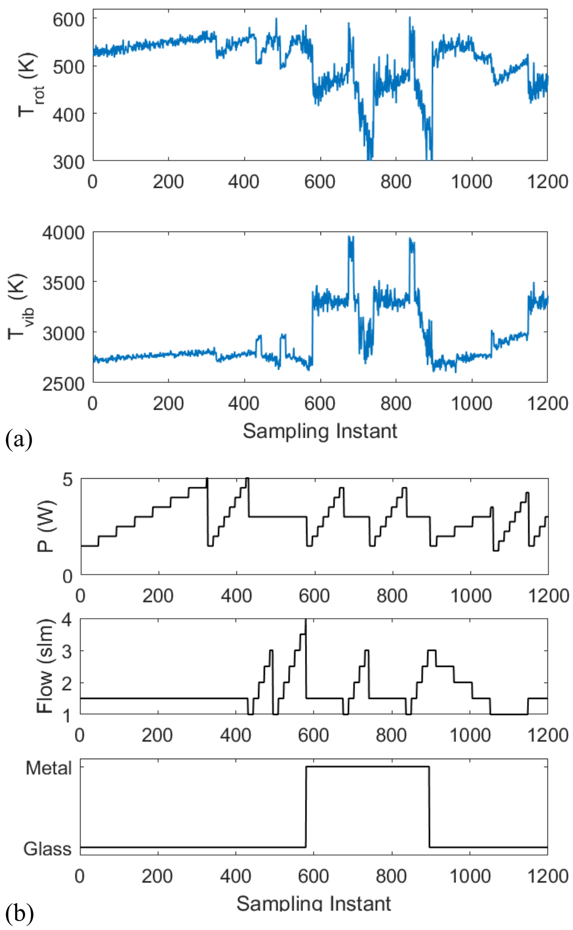

Fig. 8. Training data set collected from the APPJ setup used in linear regression and *k*-means clustering. (a) T_{rot} and T_{vib} estimates from Massive OES and (b) the corresponding operating conditions of the APPJ.

train the machine learning algorithms. Our findings suggest that, despite the simplicity of the supervised and unsupervised learning methods utilized in this work, the resulting inference quality is fairly high for the considered applications. This serves to show the potential promise of machine learning for real-time diagnosis of cold atmospheric plasmas. In particular, machine learning can create vast opportunities in plasma medicine for modeling the end effects (i.e., *dosage*) of plasma treatment, enabling development of personalized treatment protocols [41].

In future research, we will investigate the application of data analytics for building accurate data-driven models of the plasma. Such models have the capacity to be combined with real-time diagnostic tools to enable high-performance automation and process control of CAP devices.

APPENDIX
TRAINING DATA SETS

We collect OES over varying power, flow rate, and substrate types. We fit synthetic spectra using Massive OES [22] to obtain corresponding values of the outputs T_{rot} and T_{vib} for each condition. The operating conditions as well as the output variables used for training are shown in Fig. 8. We use

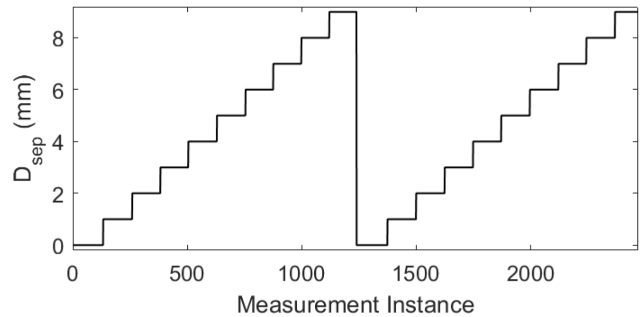

Fig. 9. Training data set collected from the plasma flashlight setup used in Gaussian process regression.

this training data for determination of T_{rot} and T_{vib} as well as substrate discrimination.

For the separation distance inference, electro-acoustic emission from the plasma flashlight is collected over a range of separation distances for which the discharge remains coupled to the ground plate. The conditions over which the training data are collected is presented in Fig. 9.

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REFERENCES

[1] M. Laroussi, "Plasma medicine: A brief introduction," *Plasma*, vol. 1, no. 1, pp. 47–60, 2018.

[2] K.-D. Weltmann and T. Von Woedtke, "Campus PlasmaMed—From basic research to clinical proof," *IEEE Trans. Plasma Sci.*, vol. 39, no. 4, pp. 1015–1025, Apr. 2011.

[3] M. G. Kong *et al.*, "Plasma medicine: An introductory review," *New J. Phys.*, vol. 11, pp. 1–35, Nov. 2009.

[4] I. Gerber *et al.*, "Time behaviour of helium atmospheric pressure plasma jet electrical and optical parameters," *Appl. Sci.*, vol. 7, no. 8, p. 812, 2017.

[5] M. Dünnbier *et al.*, "Ambient air particle transport into the effluent of a cold atmospheric-pressure argon plasma jet investigated by molecular beam mass spectrometry," *J. Phys. D Appl. Phys.*, vol. 46, no. 43, 2013, Art. no. 435203.

[6] S. A. Norberg, E. Johnsen, and M. J. Kushner, "Helium atmospheric pressure plasma jets touching dielectric and metal surfaces," *J. Appl. Phys.*, vol. 118, no. 1, pp. 1–13, 2015.

[7] S. Reuter *et al.*, "The influence of feed gas humidity versus ambient humidity on atmospheric pressure plasma jet-effluent chemistry and skin cell viability," *IEEE Trans. Plasma Sci.*, vol. 43, no. 9, pp. 3185–3192, Sep. 2015.

[8] M. Janda, Z. Machala, A. Niklová, and V. Martišovitš, "The streamer-to-spark transition in a transient spark: A DC-driven nanosecond-pulsed discharge in atmospheric air," *Plasma Sources Sci. Technol.*, vol. 21, no. 4, 2012, Art. no. 045006.

[9] E. Stoffels, Y. Sakiyama, and D. B. Graves, "Cold atmospheric plasma: Charged species and their interactions with cells and tissues," *IEEE Trans. Plasma Sci.*, vol. 36, no. 4, pp. 1441–1457, Aug. 2008.

[10] D. Dobrynin, G. Fridman, G. Friedman, and A. Fridman, "Physical and biological mechanisms of direct plasma interaction with living tissue," *New J. Phys.*, vol. 11, Nov. 2009, Art. no. 115020.

[11] D. Gidon, D. B. Graves, and A. Mesbah, "Effective dose delivery in atmospheric pressure plasma jets for plasma medicine: A model predictive control approach," *Plasma Sources Sci. Technol.*, vol. 26, no. 8, pp. 85005–85019, 2017.

[12] D. Gidon, B. Curtis, J. A. Paulson, D. B. Graves, and A. Mesbah, "Model-based feedback control of a kHz-excited atmospheric pressure plasma jet," *IEEE Trans. Rad. Plasma Med. Sci.*, vol. 2, no. 2, pp. 129–137, Mar. 2018.

[13] Y. Lyu, L. Lin, E. Gjika, T. Lee, and M. Keidar, "Mathematical modeling and control for cancer treatment with cold atmospheric plasma jet," *J. Phys. D Appl. Phys.*, vol. 52, no. 18, 2019, Art. no. 185202.

[14] D. Gidon, D. B. Graves, and A. Mesbah, "Spatial thermal dose delivery in atmospheric pressure plasma jets," *Plasma Sources Sci. Technol.*, vol. 28, no. 2, 2019, Art. no. 025006.

[15] H. F. Döbele, T. Mosbach, K. Niemi, and V. Schulz-Von Der Gathen, "Laser-induced fluorescence measurements of absolute atomic densities: Concepts and limitations," *Plasma Sources Sci. Technol.*, vol. 14, no. 2, pp. S31–S41, 2005.

[16] S. Große-Kreul *et al.*, "Mass spectrometry of atmospheric pressure plasmas," *Plasma Sources Sci. Technol.*, vol. 24, no. 4, 2015, Art. no. 044008.

[17] A. Lo, G. Cléon, P. Vervisch, and A. Cessou, "Spontaneous Raman scattering: A useful tool for investigating the afterglow of nanosecond scale discharges in air," *Appl. Phys. B Lasers Opt.*, vol. 107, no. 1, pp. 229–242, 2012.

[18] H.-R. Metelmann *et al.*, "Treating cancer with cold physical plasma: On the way to evidence-based medicine," *Contributions Plasma Phys.*, vol. 58, no. 5, pp. 415–419, 2018.

[19] C. O. Laux, T. G. Spence, C. H. Kruger, and R. N. Zare, "Optical diagnostics of atmospheric pressure air plasmas," *Plasma Sources Sci. Technol.*, vol. 12, no. 2, pp. 125–138, 2003.

[20] P. J. Bruggeman, N. Sadeghi, D. C. Schram, and V. Linss, "Gas temperature determination from rotational lines in non-equilibrium plasmas: A review," *Plasma Sources Sci. Technol.*, vol. 23, no. 2, 2014, Art. no. 023001.

[21] N. O'Connor, V. Milosavljevi, and S. Daniels, "Development of a real time monitor and multivariate method for long term diagnostics of atmospheric pressure dielectric barrier discharges: Application to He, He/N_2, and He/O_2 discharges," *Rev. Sci. Instrum.*, vol. 82, no. 8, 2011, Art. no. 083501.

[22] J. Voráč, P. Synek, L. Potočňáková, J. Hnilica, and V. Kudrle, "Batch processing of overlapping molecular spectra as a tool for spatio-temporal diagnostics of power modulated microwave plasma jet," *Plasma Sources Sci. Technol.*, vol. 26, no. 2, 2017, Art. no. 025010.

[23] Y. Wang and P. Zhao, "Noncontact acoustic analysis monitoring of plasma arc welding," *Int. J. Pressure Vessels Piping*, vol. 78, no. 1, pp. 43–47, 2001.

[24] M. Boinet, S. Verdier, S. Maximovitch, and F. Dalard, "Application of acoustic emission technique for in situ study of plasma anodising," *NDT E Int.*, vol. 37, no. 3, pp. 213–219, 2004.

[25] V. J. Law, F. T. O'Neill, and D. P. Dowling, "Evaluation of the sensitivity of electro-acoustic measurements for process monitoring and control of an atmospheric pressure plasma jet system," *Plasma Sources Sci. Technol.*, vol. 20, no. 3, 2011, Art. no. 035024.

[26] N. O'Connor and S. Daniels, "Passive acoustic diagnostics of an atmospheric pressure linear field jet including analysis in the time-frequency domain," *J. Appl. Phys.*, vol. 110, no. 1, 2011, Art. no. 013308.

[27] J. L. Walsh, F. Iza, N. B. Janson, V. J. Law, and M. G. Kong, "Three distinct modes in a cold atmospheric pressure plasma jet," *J. Phys. D Appl. Phys.*, vol. 43, no. 7, 2010, Art. no. 075201.

[28] J. J. Liu and M. G. Kong, "Sub-60 °C atmospheric helium–water plasma jets: Modes, electron heating and downstream reaction chemistry," *J. Phys. D Appl. Phys.*, vol. 44, no. 34, 2011, Art. no. 345203.

[29] V. J. Law *et al.*, "Decoding of atmospheric pressure plasma emission signals for process control," *Chaotic Model. Simulat.*, vol. 1, pp. 69–76, Jun. 2011.

[30] C. M. Bishop, *Pattern Recognition and Machine Learning* (Information Science and Statistics). Heidelberg, Germany: Springer-Verlag, 2006.

[31] K. P. Murphy, *Machine Learning*. Heidelberg, Germany: Springer-Verlag, 1991.

[32] M. A. DePristo *et al.*, "A framework for variation discovery and genotyping using next-generation DNA sequencing data," *Nat. Genet.*, vol. 43, no. 5, pp. 491–498, 2011.

[33] M. Pal and P. M. Mather, "Support vector machines for classification in remote sensing," *Int. J. Remote Sens.*, vol. 26, no. 5, pp. 1007–1011, 2005.

[34] F. Sebastiani, "Machine learning in automated text categorization," *ACM Comput. Surveys*, vol. 34, no. 1, pp. 1–47, 2002.

[35] B. Pang, L. Lee, and S. Vaithyanathan, "Thumbs up? Sentiment classification using machine learning techniques," in *Proc. Conf. Empirical Methods Natural Lang. Process. (EMNLP)*, 2002, pp. 79–86.

[36] P. Duygulu, K. Barnard, J. F. G. de Freitas, and D. A. Forsyth, "Object recognition as machine translation: Learning a lexicon for a fixed image vocabulary," in *Proc. Eur. Conf. Comput. Vis.*, 2002, pp. 97–112.

[37] K. Irani, J. Cheng, U. Fayyad, and Z. Qian, "Applying machine learning to semiconductor manufacturing," *IEEE Expert*, vol. 8, no. 1, pp. 41–47, Feb. 1993.

[38] B. Kim and S. Park, "An optimal neural network plasma model: A case study," *Chemometrics Intell. Lab. Syst.*, vol. 56, no. 1, pp. 39–50, 2001.

[39] P.-H. Chou, M.-J. Wu, and K.-K. Chen, "Integrating support vector machine and genetic algorithm to implement dynamic wafer quality prediction system," *Expert Syst. Appl.*, vol. 37, no. 6, pp. 4413–4424, 2010.

[40] G. Erten, A. Gharbi, F. Salam, T. Grotjohn, and J. Asmussen, "Using neural networks to control the process of plasma etching and deposition," in *Proc. IEEE Int. Conf. Neural Netw. Conf.*, vol. 2. Washington, DC, USA, 1996, pp. 1091–1096.

[41] R. Yang and R. Chen, "Real-time plasma process condition sensing and abnormal process detection," *Sensors*, vol. 10, no. 6, pp. 5703–5723, 2010.

[42] A. Mesbah and D. B. Graves, "Machine learning for modeling, diagnostics, and control of non-equilibrium plasmas," *J. Phys. D Appl. Phys.*, 2019.

[43] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, Jan. 2011.

[44] M. Abadi *et al.*, "TensorFlow: A system for large-scale machine learning," in *Proc. 12th USENIX Symp. Oper. Syst. Design Implement. (OSDI)*, Savannah, GA, USA, 2016, pp. 265–284.

[45] A. C. Mueller and S. Guido, *Introduction to Machine Learning With Python*, 1st ed. Sebastopol, CA, USA: O'Reilly Media, 2016.

[46] Y. Tsai *et al.*, "Pattern and knowledge extraction using process data analytics: A tutorial," in *Proc. 10th IFAC Int. Symp. Adv. Control Chem. Process.*, Shenyang, China, 2018, pp. 13–18.

[47] J. T. McClave and T. Sincich, *Statistics*. Upper Saddle River, NJ, USA: Prentice-Hall, 2009.

[48] J. Friedman, T. Hastie, and R. Tibshirani, "Regularization paths for generalized linear models via coordinate descent," *J. Stat. Softw.*, vol. 33, no. 1, pp. 1–22, 2010.

[49] D. Arthur and S. Vassilvitskii, "K-Means++: The advantages of careful seeding," in *Proc. 18th Annu. ACM-SIAM Symp. Discr. Algorithms*, vol. 8. New Orleans, LA, USA, 2007, pp. 1027–1025.

[50] C. E. Rasmussen and C. K. I. Williams, *Gaussian Processes for Machine Learning*. Cambridge, MA, USA: MIT Press, 2004.

[51] X. Pei *et al.*, "Inactivation of a 25.5 μm enterococcus faecalis biofilm by a room-temperature, battery-operated, handheld air plasma jet," *J. Phys. D Appl. Phys.*, vol. 45, no. 16, 2012, Art. no. 165205.

[52] Open Music Labs. (Dec. 2015). *ArduinoFFT—Fast Fourier Transform Library*. Accessed: Aug. 23, 2018. [Online]. Available: http://wiki.openmusiclabs.com/wiki/ArduinoFFT

[53] X. Lu and K. K. Ostrikov, "Guided ionization waves: The physics of repeatability," *Appl. Phys. Rev.*, vol. 5, no. 3, 2018, Art. no. 031102.

[54] D. Spether *et al.*, "Real-time tissue differentiation based on optical emission spectroscopy for guided electrosurgical tumor resection," *Biomed. Opt. Exp.*, vol. 6, no. 4, pp. 1419–1428, 2015.

[55] K.-D. Weltmann *et al.*, "Atmospheric pressure plasma jet for medical therapy: Plasma parameters and risk estimation," *Contributions Plasma Phys.*, vol. 49, no. 9, pp. 631–640, 2009.

[56] G. Bennet, F. Blanes, J. E. Simo, and P. Perez, "Using infrared sensors for distance measurement in mobile robots," *Robot. Auton. Syst.*, vol. 40, no. 4, pp. 255–266, 2002.

[57] M. Kelemen *et al.*, "Distance measurement via using of ultrasonic sensor," *J. Autom. Control*, vol. 3, no. 3, pp. 71–74, 2015.

[58] M. Lindner, I. Schiller, A. Kolb, and R. Koch, "Time-of-flight sensor calibration for accurate range sensing," *Comput. Vis. Image Understand.*, vol. 114, no. 12, pp. 1318–1328, 2010.

[59] D. B. Graves, "Mechanisms of plasma medicine: Coupling plasma physics, biochemistry, and biology," *IEEE Trans. Radiat. Plasma Med. Sci.*, vol. 1, no. 4, pp. 281–292, Jul. 2017.